`GeoModels` Tutorial: simulation, estimation and prediction of spatial count data using Poisson random fields

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Introduction

In this tutorial we show how to analyze spatial count data using Poisson random fields as proposed in Morales-Navarrete et al. [2024], i.e., random fields with Poisson marginal distribution, using the R package `GeoModels` [Bevilacqua et al., 2026].

We first load the R libraries needed in this tutorial and set the name of the model in the `GeoModels` package.

```
rm(list=ls());
require(GeoModels);
require(fields);
model="Poisson"; # model name in the GeoModels package
```

Poisson random fields

The definition of a Poisson random field starts by considering a 'parent' Gaussian random field $G = \{G(\boldsymbol{s}), \boldsymbol{s} \in A\}$, where $\boldsymbol{s}$ represents a location in the domain A. In this tutorial we consider the spatial case, i.e., $A \subseteq \mathbb{R}^2$. However, the package `GeoModels` also allows one to work with spatio-temporal data or data defined on a sphere of arbitrary radius. The Gaussian random field G is assumed weakly stationary with zero mean, unit variance and correlation function $\rho(\boldsymbol{h}) = \text{cor}\{G(\boldsymbol{s}+\boldsymbol{h}), G(\boldsymbol{s})\}$.

Let G_1, G_2 be two independent copies of G and let us define the random field $W = \{W(\boldsymbol{s}), \boldsymbol{s} \in A\}$ as:

$$W(\boldsymbol{s}) := \frac{1}{2\lambda(\boldsymbol{s})} \sum_{k=1}^{2} G_k(\boldsymbol{s})^2, \tag{1}$$

where $\lambda(\boldsymbol{s}) > 0$ is a non-random function. It turns out that $W(\boldsymbol{s})$ has a marginal exponential distribution with rate parameter $\lambda(\boldsymbol{s})$, i.e., $W(\boldsymbol{s}) \sim \text{Exp}(\lambda(\boldsymbol{s}))$, with $\text{E}\{W(\boldsymbol{s})\} = 1/\lambda(\boldsymbol{s})$. The random field W is stationary when $\lambda(\boldsymbol{s})$ is constant.

By considering an infinite sequence of independent copies $W_1, W_2, \ldots$ of W, a Poisson random field $N_t := \{N_t(\boldsymbol{s}), \boldsymbol{s} \in A\}$ for $t > 0$ can be defined as:

$$N_t(\boldsymbol{s}) := \begin{cases} 0, & \text{if } 0 \le t < S_1(\boldsymbol{s}), \\ \max\{n \ge 1 : S_n(\boldsymbol{s}) \le t\}, & \text{if } S_1(\boldsymbol{s}) \le t, \end{cases} \tag{2}$$

where $S_n(\boldsymbol{s}) = \sum_{i=1}^{n} W_i(\boldsymbol{s})$ is the sum of n inter-arrival times. $N_t(\boldsymbol{s})$ represents the random total number of events that have occurred up to time t at location site $\boldsymbol{s}$; that is, for each $\boldsymbol{s}$

it is a Poisson counting process. Given two spatial locations $\boldsymbol{s}_1, \boldsymbol{s}_2$, the associated Poisson counting processes $N_t(\boldsymbol{s}_1)$ and $N_t(\boldsymbol{s}_2)$ can be correlated.

The previous model can be viewed as a spatial generalization of the Poisson counting process [Cox, 1970, Mainardi et al., 2007], where we consider independent copies of positive random fields as 'inter-arrival times' instead of an i.i.d. sequence of positive random variables. By construction, for each $\boldsymbol{s}$, $N_t(\boldsymbol{s}) \sim \text{Pois}(t\lambda(\boldsymbol{s}))$. For the purpose of this tutorial we take $t = 1$ and set $N := N_1$. Then $\text{E}\{N(\boldsymbol{s})\} = \text{Var}\{N(\boldsymbol{s})\} = \lambda(\boldsymbol{s})$, and the correlation function of the non-stationary Poisson random field is given by Morales-Navarrete et al. [2024]:

$$\rho_N(\boldsymbol{s}_i, \boldsymbol{s}_j) = \frac{\rho^2(\boldsymbol{h})[1-\rho^2(\boldsymbol{h})]}{\sqrt{\lambda(\boldsymbol{s}_i)\lambda(\boldsymbol{s}_j)}} \sum_{r=0}^{\infty} \gamma^* \left(r+1, \frac{\lambda(\boldsymbol{s}_i)}{1-\rho^2(\boldsymbol{h})}\right) \gamma^* \left(r+1, \frac{\lambda(\boldsymbol{s}_j)}{1-\rho^2(\boldsymbol{h})}\right), \tag{3}$$

with $\boldsymbol{h} = \boldsymbol{s}_i - \boldsymbol{s}_j$, and $\gamma^*(\cdot,\cdot)$ the regularized lower incomplete gamma function (with the diagonal value defined by continuity).

Simulation of Poisson random fields

We first set the spatial coordinates (Figure 1):

```
set.seed(1989);
N=500;
coords=cbind(runif(N),runif(N));
plot(coords,pch=20,xlab="",ylab="");
```

Stationary Poisson random fields

Let us assume that $\lambda(\boldsymbol{s}) = \lambda$, i.e., we are assuming a constant mean for the Poisson random field. In this case correlation (3) simplifies to

$$\rho_N(\boldsymbol{h}, \lambda) = \rho^2(\boldsymbol{h})\Big[1 - \exp\{-z(\boldsymbol{h}, \lambda)\}\{I_0(z(\boldsymbol{h}, \lambda)) + I_1(z(\boldsymbol{h}, \lambda))\}\Big], \tag{4}$$

where $z(\boldsymbol{h}, \lambda) = 2\lambda/[1-\rho^2(\boldsymbol{h})]$, and $I_a(\cdot)$ denotes the modified Bessel function of the first kind of order a. To obtain a simulation from a stationary Poisson random field we need to specify the mean and a parametric correlation model $\rho(\boldsymbol{h})$ for the underlying Gaussian random field.

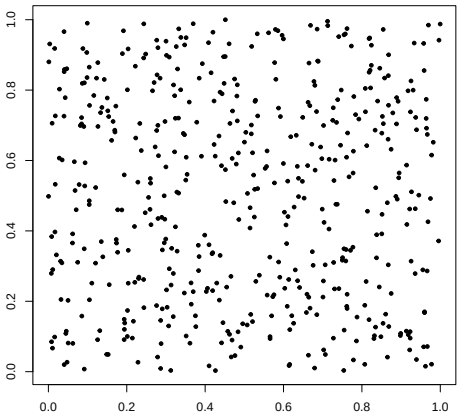


Figure 1: Spatial location sites used in the tutorial.

For the correlation function $\rho(\boldsymbol{h})$ of the 'parent' Gaussian random field G we assume an isotropic Matérn model [Matérn, 1986]:

$$\rho_{\alpha,\gamma}(\boldsymbol{h}) = \frac{2^{1-\gamma}}{\Gamma(\gamma)} \left(\frac{\|\boldsymbol{h}\|}{\alpha} \right)^{\gamma} K_{\gamma} \left(\frac{\|\boldsymbol{h}\|}{\alpha} \right), \qquad \|\boldsymbol{h}\| \geq 0, \tag{5}$$

where K_γ is a modified Bessel function of the second kind of order γ, $\gamma > 0$ is the smoothness parameter and $\alpha > 0$ the spatial scale parameter. Then we set the parameters associated with this correlation model:

```
corrmodel="Matern"; ## correlation model
scale=0.25/3; ## scale parameter
smooth=0.5; ## smooth parameter
nugget=0; ## nugget parameter
```

Finally we set the regression parameter β in $\lambda = e^{\beta}$:

```
mean=1.5; # regression parameter (log-mean)
```

Simulation uses the stochastic construction based on the underlying Gaussian random fields. We are now ready to simulate a realization of the Poisson random field N using the function `GeoSim`:

```
param=list(nugget=nugget,mean=mean,scale=scale,
           smooth=smooth);
data_s <- GeoSim(coordx=coords,corrmodel=corrmodel,
                          param=param,model=model)$data;
```

Note that empirical mean and variance and theoretical mean are very close in this realization:

```
mean(data_s); var(data_s)
[1] 4.516
[1] 4.402549
exp(mean)
[1] 4.481689
```

The following figure shows the distribution of the data:

```
plot(table(data_s),ylab="Frequency")
```

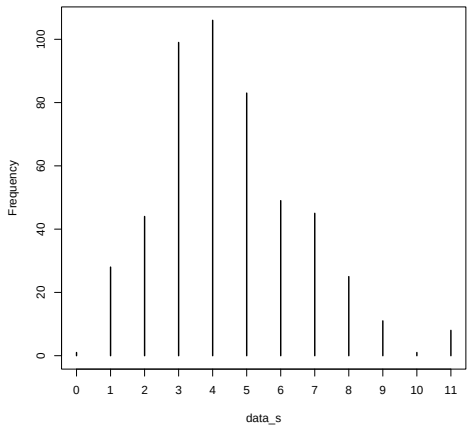


Figure 2: Distribution of the simulated Poisson random field.

Non Stationary Poisson random fields

A non-stationary Poisson random field can be specified by assuming $\lambda(\boldsymbol{s}) = \exp\{X(\boldsymbol{s})^T\boldsymbol{\beta}\}$, where $X(\boldsymbol{s})$ is a k-dimensional vector of covariates and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_k)^T$ is a k-dimensional vector of (unknown) parameters. In this tutorial we assume $k = 2$.

Thus, to obtain a realization from a non-stationary Poisson random field we need to specify the regression parameters and a parametric correlation model $\rho(\boldsymbol{h})$ for the underlying Gaussian random field G. For the correlation function $\rho(\boldsymbol{h})$ we assume a special case

of the isotropic Generalized Wendland class [Bevilacqua et al., 2019]:

$$\rho_{\alpha,\delta}(\boldsymbol{h}) := \begin{cases} (1 - \|\boldsymbol{h}\|/\alpha)^{\delta}, & \|\boldsymbol{h}\| < \alpha, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Then we set the parameters associated with this correlation model:

```
corrmodel="Wend0"; ## correlation model
scale=0.2; ## scale parameter
power2=4; ## power parameter
nugget=0; ## nugget parameter
```

Finally we set the regression parameters and the regression matrix:

```
mean=1.5; # regression parameter beta_0
mean1=-0.25; # regression parameter beta_1
a0=rep(1,N); a1=runif(N);
X=cbind(a0,a1); ## regression matrix
```

To simulate a realization of the Poisson random field N we use the function `GeoSim`:

```
param=list(nugget=nugget,mean=mean,mean1=mean1,
          scale=scale,power2=power2);
data_ns <- GeoSim(coordx=coords,corrmodel=corrmodel,
                            param=param,X=X,model=model)$data;
```

Estimation of Poisson random fields

Estimation of regression and correlation parameters of the Poisson random field N can be performed using pairwise likelihood estimation. Let $\Pr\{N(\boldsymbol{s}_i) = n_i, N(\boldsymbol{s}_j) = n_j\}$ be the probability mass function of the bivariate random vector $(N(\boldsymbol{s}_i), N(\boldsymbol{s}_j))^T$ given in Morales-Navarrete et al. [2024].

Given a partial realization $(n(\boldsymbol{s}_1), \ldots, n(\boldsymbol{s}_l))^T$ of the Poisson random field N defined in equation (2), the pairwise log-likelihood function is defined as:

$$pl(\boldsymbol{\theta}) = \sum_{i=1}^{l} \sum_{\substack{j=1 \\ j \neq i}}^{l} \log[\Pr\{N(\boldsymbol{s}_i) = n_i, N(\boldsymbol{s}_j) = n_j\}]\, c_{ij}, \tag{7}$$

where an efficient way to specify the non-symmetric weights, from a computational viewpoint, is based on neighborhoods:

$$c_{ij}(k) = \begin{cases} 1, & \boldsymbol{s}_i \in N_k(\boldsymbol{s}_j), \\ 0, & \text{otherwise}, \end{cases} \tag{8}$$

where $N_k(\boldsymbol{s}_l)$ is the set of the neighbors of order $k = 1, 2, \ldots$ of the point $\boldsymbol{s}_l$. The pairwise likelihood estimator $\widehat{\boldsymbol{\theta}}_{pl}$ is obtained maximizing (7) with respect to $\boldsymbol{\theta}$. In the `GeoModels` package we can choose the fixed parameters and the parameters that can be estimated. Pairwise likelihood estimation can be performed using the function `GeoFit`. In this example, we use the box-constrained optimizer `nlminb`.

Stationary Case

```
optimizer="nlminb";
fixed1<-list(nugget=0,smooth=0.5);
start1<-list(mean=1.5,scale=0.25/3);
lower<-list(mean=-5,scale=1e-6);
upper<-list(mean=5,scale=2);
neighb=3;
corrmodel="Matern";
fit1 <- GeoFit(data=data_s,coordx=coords,corrmodel=corrmodel,
 optimizer=optimizer,lower=lower,upper=upper,neighb=neighb,
 start=start1,fixed=fixed1,model=model);
```

Note that `neighb=3` specifies the number of neighbors used by the weight function (8), i.e., $k = 3$. The following output from the original tutorial illustrates the pairwise likelihood estimate:

```
fit1
##########################################################
Maximum Composite-Likelihood Fitting of Poisson Random Fields
Setting: Marginal Composite-Likelihood
Model: Poisson
Type of the likelihood objects: Pairwise
Covariance model: Matern
Optimizer: nlminb
Number of spatial coordinates: 500
```

```
Number of estimated parameters: 2
Type of convergence: Successful
Maximum log-Composite-Likelihood value: -6189.99
Estimated parameters:
   mean scale
 1.51398 0.07921
############################################################
```

An alternative, less efficient and computationally easier estimator can be obtained by assuming a misspecified Gaussian model in the pairwise likelihood estimation method. Specifically, if in the estimation step we assume a Gaussian random field with the same mean, variance and correlation function of the Poisson random field (see Morales-Navarrete et al., 2024), then a misspecified Gaussian pairwise likelihood estimator is available by changing the name of the model in `GeoFit`:

```
fit2 <- GeoFit(data=data_s,coordx=coords,corrmodel=corrmodel,
 optimizer=optimizer,lower=lower,upper=upper,neighb=neighb,
 start=start1,fixed=fixed1,model="Gaussian_misp_Poisson");
```

The two estimates are quite similar in this case, but in general the misspecified Gaussian assumption can lead to a loss of efficiency, which increases with decreasing marginal mean [Morales-Navarrete et al., 2024].

```
unlist(fit1$param)
     mean scale
1.51398129 0.07921077
unlist(fit2$param)
     mean scale
1.51140565 0.06933799
```

The empirical and estimated semi-variograms can be compared with `GeoCovariogram`:

```
vario <- GeoVariogram(data=data_s,coordx=coords,maxdist=0.4)
GeoCovariogram(fit1,show.vario=TRUE,vario=vario,pch=20)
```

Non Stationary Case

```
optimizer="nlminb";
corrmodel="Wend0";
```

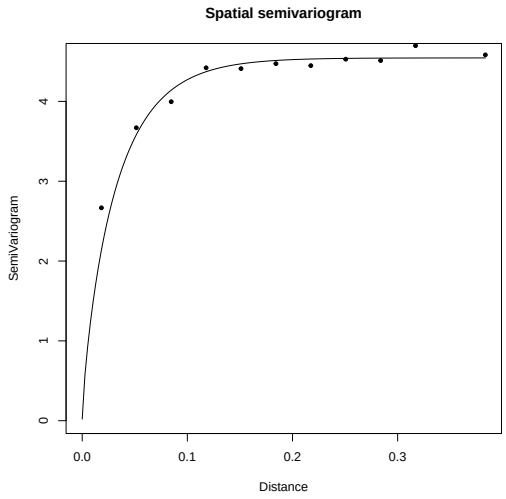


Figure 3: Empirical and estimated semi-variograms for Poisson data.

```
fixed2<-list(nugget=0,power2=4);
start2<-list(mean=1.5,mean1=-0.25,scale=0.2);
lower<-list(mean=-5,mean1=-5,scale=1e-6);
upper<-list(mean=5,mean1=5,scale=2);
fit1_ns <- GeoFit(data=data_ns,coordx=coords,corrmodel=corrmodel,
 optimizer=optimizer,lower=lower,upper=upper,X=X,
 neighb=neighb,start=start2,fixed=fixed2,model=model);
```

The following output is reproduced from the original tutorial:

```
fit1_ns
##########################################################
Maximum Composite-Likelihood Fitting of Poisson Random Fields
Setting: Marginal Composite-Likelihood
Model: Poisson
Type of the likelihood objects: Pairwise
Covariance model: Wend0
Optimizer: nlminb
Number of spatial coordinates: 500
Number of estimated parameters: 3
Type of convergence: Successful
Maximum log-Composite-Likelihood value: -6433.92
Estimated parameters:
```

```
   mean mean1 scale
 1.5531 -0.2445 0.1476
##########################################################
```

The misspecified Gaussian pairwise likelihood estimate can be obtained by changing the model name in `GeoFit`:

```
fit2_ns <- GeoFit(data=data_ns,coordx=coords,
 corrmodel=corrmodel,optimizer=optimizer,
 lower=lower,upper=upper,X=X,neighb=neighb,
 start=start2,fixed=fixed2,model="Gaussian_misp_Poisson");
```

The two estimates are quite similar in this case, but in general the misspecified Gaussian assumption can lead to a loss of efficiency.

```
unlist(fit1_ns$param)
     mean mean1 scale
 1.5530974 -0.2444645 0.1475900
unlist(fit2_ns$param)
     mean mean1 scale
 1.5603522 -0.2369685 0.1422761
```

Prediction of Poisson random fields

For a given spatial location $\boldsymbol{s}_0$ with associated covariates $X(\boldsymbol{s}_0)$, the optimal linear predictor of a Poisson random field is given by:

$$\widehat{N}(\boldsymbol{s}_0) = \lambda(\boldsymbol{s}_0) + \boldsymbol{c}^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{N} - \boldsymbol{\lambda}), \tag{9}$$

where $\boldsymbol{\lambda} = (\lambda(\boldsymbol{s}_1), \ldots, \lambda(\boldsymbol{s}_l))^T$, $\boldsymbol{c} = \left(\sqrt{\lambda(\boldsymbol{s}_0)\lambda(\boldsymbol{s}_i)}\rho_N(\boldsymbol{s}_0, \boldsymbol{s}_i)\right)_{i=1}^{l}$, and

$$\boldsymbol{\Sigma} = \left(\sqrt{\boldsymbol{\lambda}\boldsymbol{\lambda}^T}\right) \odot \left[\rho_N(\boldsymbol{s}_i, \boldsymbol{s}_j)\right]_{i,j=1}^{l},$$

where $\odot$ is the Schur (Hadamard) product and the square root is applied elementwise. The associated mean squared error is given by:

$$\mathrm{MSE}\{\widehat{N}(\boldsymbol{s}_0)\} = \lambda(\boldsymbol{s}_0) - \boldsymbol{c}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{c}. \tag{10}$$

The predictor is the best linear predictor based on covariance function (3). If the parameters are unknown, both (9) and (10) can be computed by replacing the parameters with the

pairwise likelihood estimates. Kriging and associated MSE can be obtained using the `GeoKrig` function.

Stationary Case

We first specify the spatial locations at which to predict, considering a spatial regular grid:

```
xx=seq(0,1,0.015)
loc_to_pred=as.matrix(expand.grid(xx,xx))
```

Then the optimal linear prediction (9) using the estimated parameters can be performed by the `GeoKrig` function (computation can be time consuming):

```
pr=GeoKrig(fit1,loc=loc_to_pred,mse=TRUE)
```

Finally, a kriging map with associated mean square error (Figure 4) can be obtained with the following code:

```
quilt.plot(coords[,1],coords[,2],data_s,main="Data")
map=matrix(pr$pred,ncol=length(xx))
image.plot(xx,xx,map,xlab="",ylab="",main="Kriging")
map_mse=matrix(pr$mse,ncol=length(xx))
image.plot(xx,xx,map_mse,xlab="",ylab="",main="MSE")
```

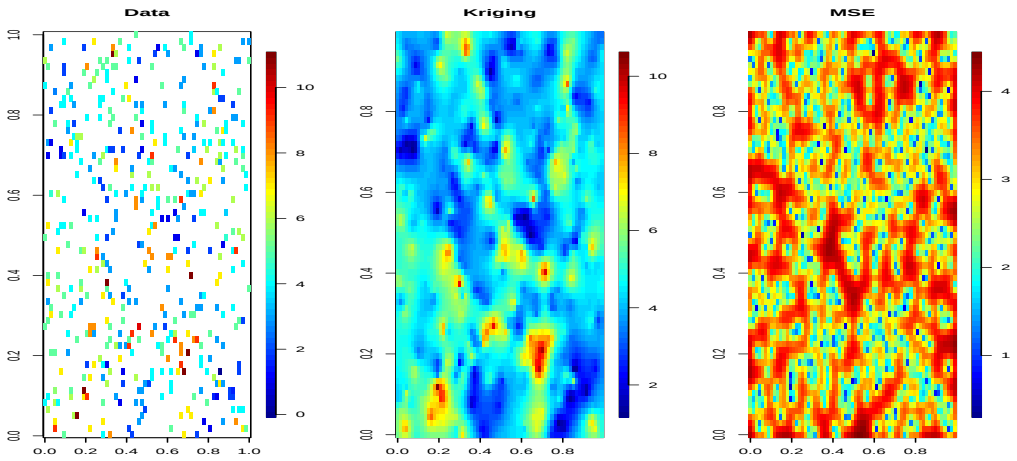


Figure 4: From left to right: observed spatial data, associated kriging map and mean square error map.

Non Stationary Case

We need to specify the spatial locations and covariates at prediction locations. In this example we consider the same spatial regular grid:

```
set.seed(609)
NN=nrow(loc_to_pred)
a0=rep(1,NN); a1=runif(NN)
Xloc=cbind(a0,a1)
```

Then the linear prediction (9) using the estimated parameters is performed using the `GeoKrig` function:

```
pr_ns=GeoKrig(fit1_ns,loc=loc_to_pred,Xloc=Xloc,mse=TRUE)
```

Finally, a kriging map with associated MSE (Figure 5) is obtained:

```
quilt.plot(coords[,1],coords[,2],data_ns,main="Data")
map_ns=matrix(pr_ns$pred,ncol=length(xx))
image.plot(xx,xx,map_ns,xlab="",ylab="",main="Kriging")
map_mse_ns=matrix(pr_ns$mse,ncol=length(xx))
image.plot(xx,xx,map_mse_ns,xlab="",ylab="",main="MSE")
```

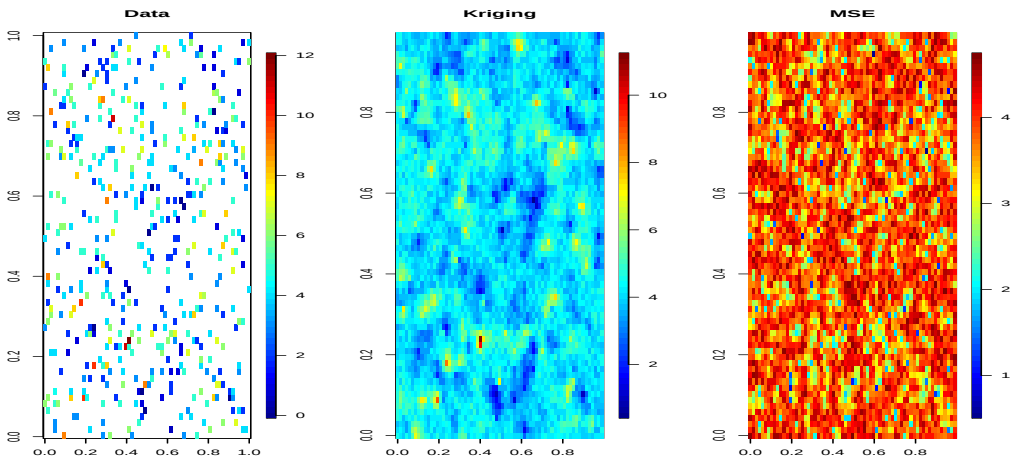


Figure 5: From left to right: observed spatial data, associated kriging map and mean square error map.

References

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